REBECCA WILLIAMS

Professor of Volcanology
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About me

I'm a volcanologist who takes an interdisciplinary approach to volcano research. My research focusses on volcano-stratigraphy and hazardous volcanic flows, such as pyroclastic density currents and lahars. I have an interest in social-historical volcanology, and the communication of volcano science. All this drives towards understanding how communities may be affected by volcanic eruptions past, present and future.

I'm a passionate educator with a particular focus on equality, diversity and inclusivity and curriculum design. My research in this area seeks to understand the links between the colonial legacy of Earth Science, the pedagogy of our discipline and inequities in the access, retention and success of diverse student cohorts. My practice in this arena seeks to effect positive change in Higher Education, with a particular focus on the Earth Sciences across the sector, and across disciplines and faculties within my own institution.

I am a published book author and award-winning science communicator, appearing on TV, radio and news media. This work links people to places, landscapes and science through geology and geoheritage.

Education

PhD Geology | 2006 – 2010 | University of Leicester

Emplacement of radial pyroclastic density currents over irregular topography: the chemically-zoned, low aspect-ratio Green Tuff Ignimbrite, Pantelleria, Italy. (NER/S/A/2006/14156). Advisor: Prof. Michael J. Branney.

MS Geology | 2004 – 2006 | University at Buffalo, USA

Modeling lahars using Titan2D for the southern drainage of Volcán Cotopaxi: Impact on the city of Latacunga. Advisor: Prof. Michael Sheridan.

BSc Geology | 1999 – 2002| Royal Holloway university of london

Geological History of the Digne Region, France. Advisor: Dr. John Wright

- Upper 2nd Class Honours

Professional Qualifications & Fellowships

Qualifications

- 2015: Postgraduate Certificate of Academic Practise, University of Hull

Fellowships

- 2018: Senior Fellow of the Higher Education Authority
- 2015: Fellow of the Higher Education Authority

Experience

Professor of Volcanology | Loughborough University | 06/2026 – present

Reader In Volcanology | University Of Hull | 08/2022 – 306/2026

- Director of Research, School of Environmental and Life Sciences
- School of Environmental and Life Sciences Executive and Leadership Team.
- Open Research Champion

Associate Dean Student Experience | University of Hull | 03/2019 – 03/2023

- Faculty of Science & Engineering Executive and Leadership Team.
- Vice-Chair University Student Experience Strategy Group; Member of Senate and University Student Experience, Engagement and Employability Committee; Co-Chair of Faculty Education & Student Experience Committee; Chair of Mitigating Circumstances Committee.

Senior Lecturer | University of Hull | 08/2018 – 08/2022

- 2016-2018: Head of Geology Subject Group

Lecturer | University of Hull | 02/2013 – 08/2018

- 2014-2016: Deputy Learning and Teaching Director in the Department of Geography, Environment and Earth Sciences.
- 2013-2017: Programme Lead for the Geosciences degree programmes (BSc Geology and BSc Geology with Physical Geography).

Teaching Fellow | University of Leicester | 10/2010 – 02/2013

- 0.6 FTE
- UCAS Officer
- 3rd Year coordinator

PDRA | University of Leicester | 05/2011 – 12/2011

- 0.4 FTE
- IODP Expedition 330: assessing the mantle plume component in the Louisville Seamount Trail, SW Pacific Ocean

Igneous Petrologist/Volcanologist | IODP Expedition 330 | 12/2010 – 02/2011

- Igneous Team Leader

Postgraduate demonstrator | University of Leicester | 10/2006 – 05/2010

Teaching Assistant | University at Buffalo, USA | 08/2004 – 05/2006

Dive Supervisor | Hydroactive Dive Centre | 04/2003 – 08/2004

- PADI Divemaster
- Store Manager

Assistant Gas Geochemist | Hawaiian Volcano Observatory | 07/2002 – 01/2003

Career breaks

16/06/2020 – 15/09/2021 Maternity Leave

01/03/2017 – 01/01/2019 Maternity Leave

External positions held

- 2026: External assessor, 2nd Level geology module, Open University
- Since 2023: NERC PRC Member
- Since 2023: Co-Chair Commission for Volcanogenic Sediments, IAVCEI.
- Since 2022: NERC Pushing the Frontiers Panel Member
- 2022: NERC IRF Panel Member
- Since 2017: Executive Board for University Geoscience UK (nominated). Head of Employability Portfolio (2017-2020). Head of Equality, Diversity and Inclusivity Portfolio (2020 – present).
- 2021: QAA Earth Science, Environmental Sciences and Environmental Studies Subject Benchmark Statement Advisory Board. Chair of Equality, Diversity and Inclusivity Working Group.
- External Examiner: 2024 – present, BSc Earth Science, University of Portsmouth; 2019-2024, BSc Geology and MSci Geoscience, University Keele.
- Examiner for MSc by research and PhD theses (e.g. University of East Anglia, University of Durham, Lancaster University).

Awards

External

- 2017: British Science Association Charles Lyell Award for ""cutting-edge research and exceptional ability to communicate work to wider audiences".
- 2008: Shell Professional Development Award.

Internal

- 2025: University of Hull Research and Knowledge Exchange Award for: "Equality, Diversity, and Inclusion Excellence in Research and Knowledge Exchange Award".
- 2018: Faculty of Science Learning and Teaching Award for "Outstanding Performance by a Team Across Multiple Categories" to "Geology Teaching and Support Staff led by Rebecca Williams"
- 2009: P.C. Silvester-Bradley Award, Department of Geology, Leicester. Best PhD student research seminar.
- 2006 : Reginald H. Pegrum Graduate Student Award for Excellence in Research, University at Buffalo.

Research Experience

Leadership

- PI of the Catastrophic Flows Research Cluster
- Founder of the VOICES (Volcanoes In Communication, Engagement & Society) Group.
- Lead GeoCoLab, 2021 NERC Digital Technologies to open up environmental sciences Hackathon.
- Co-I EQUATOR Research Group

Grants

PERIOD	TITLE	FUNDER	VALUE	ROLE
2026	VolcaniCity	The Curry Fund of the Geologists' Association	£2531.43	Co-L
2025	HERC Enhancing diversity in the academic 'marzipan layer'	HERC/Hull	£8841	Co-L
2024-2025	International Ocean Discovery Program Expedition 396 Moratorium Award	NERC/IODP	£15,371	PI
2023-2025	FIAMME: (An international collaboration for a) Framework for Ignimbrite Analysis Methodologies for Modelling and hazard Evaluation, NE/Y003306/1.	NERC	£99,971	PI
2023	Extension of Decolonising UK Earth Science Pedagogy	British Geological Survey	£15K	PI
2022	SmartLapilli, Royal Society International Exchange IES\R2\222171	Royal Society	£3K	PI
2022	SmartLapilli/IAVCEI Conference Carers Support BSG-2022-20	British Society for Geomorphology	£500	PI
2022-2023	Decolonising UK Earth Science pedagogy - from the hidden histories of our geological institutions to inclusive curricula, AH/W008726/1	AHRC-NERC	£124,931	PI
2022	EQUATOR - Building solid ground for racial diversity in Geography, Earth & Environmental Sciences postgraduate research, 2021EDIE004Dowey	NERC	£96,416	Co-I
2021	GeoCoLab. NE/W007622/1	NERC	£49,991	PI
2021	A Roadmap to Strengthen Geoscience Education for Sustainable Development in Kenya	QR-GCRF	£30,080	PI
2016	Augmented Reality Geomorphology	British Society for Geomorphology	£1000	PI
2014	Lahars as landscape modifiers: how repeated lahar emplacement affects flow behaviour.	British Society for Geomorphology	£4000	PI
2014	Do humid phases in coastal and southern Libya occur synchronously? IP-1385-1113	NIGFSC	£18,000	Co-I
2016	Catastrophic Flows Research Cluster.	University of Hull	~£250,000	PI
2016	Laser ablation QQQ-ICP-MS New Initiatives Bid	University of Hull	~£378,480	Co-I

2013	HUGE (Hull University GEochemistry)	Ferens Education Trust	£400	Co-I
2011	A high-precision Nd and Hf isotope study of the Louisville Seamount Trail (IODP Expedition 330). IP-1245-0511	NIGFSC	£42,600	Res-I
2011	IODP Expedition 330: assessing the mantle plume component in the Louisville Seamount Trail, SW Pacific Ocean. NE/J005401/1	NERC	£19,462	Res-I
2009	Tracking compositional changes in volcanic glass and phenocrysts in a zoned ignimbrite using LA ICP-MS: detailed chemistry of the Green Tuff Ignimbrite, Pantelleria	John Whittaker Award University of Leicester	£300	PI
2009	For fieldwork associated with PhD research	Gill Harwood fund British Sedimentological Research Group	£300	PI
2006	Modeling Lahars Using Titan2D for the Southern Drainage of Volcán Cotopaxi: Impact on the City of Latacunga, Ecuador	NSF	$5000	Res-S
2005	For field research in Ecuador	NSF	$2000	Res-S

Project Outcomes and Deliverables

Datasets

- Brown, J., **Williams, R**., Ogburn, S., Brand, B., Breard, E. C. P., Charbonnier, S., Dowey, N., Dufek, J., Jellinek, M., Kueppers, U., Lube, G., & Rowley, P. (2026). A database of pyroclastic density current deposit characteristics compiled from a literature review (PDCD-Dat v.2026) [Data set]. NERC EDS National Geoscience Data Centre. https://doi.org/10.5285/04c9ea44-4388-4239-b31e-47566ee5fd37

Workshops

- 2026: Recording and interpreting pyroclastic stratigraphies, VMSG Annual Meeting, National Oceanographic Centre, Southampton, UK. **Organiser**
- 2023: Decolonising Earth Science Workshop, British Geological Survey, Keyworth, UK. **Organiser**

Open Educational Materials

- 2023 - Present: Infographics and videos hosted on DecolEarthSci website: https://www.decolearthsci.com/

Public Engagement Events

- 2026: Tempest and Volcanoes: Ten Things You Should Know, Talk, Festival of Ideas, York, UK.
- 2026: Tempest's Volcano Trail, Guided Walk, Festival of Ideas, York, UK.

Publications

Pre-prints

- Smith, G., **Williams, R.,** Rowley, P., Parson, D., (2023). Characterising the flow-boundary zone in fluidized granular currents. Pre-print at EarthArXiv https://doi.org/10.31223/X56958
- Rowley, P., Giordano, G., Silleni, A., Smith, G., Trolese, M., **Williams, R.,** (2023). Stationary surface waves and antidunes in dense pyroclastic density currents. Pre-print at EarthArXiv https://eartharxiv.org/repository/view/5209/
- Chen, Y., **Williams, R**., Simmons, S.M., Cartigny, M.J.B., Heijnen M.S., Parsons, D.R., Hughes Clarke, J.E., Stacey, C.D., Hage, S., Talling, P.J., Pope, E., Azpiroz-Zabala, M., Clare, M.A., Heerema, C.J., Hizzett, J.L., Hunt, J.E., Lintern, D.G., Sumner, E.J., Vellinga, A.J. and Vendettuoli, D., (2021). The Formation and Evolution of Submarine Headless Channels. Pre-print at EarthArXiv https://doi.org/10.31223/X5KD0T

Reports

- Dowey, N., Giles, S., Jackson, C A-L, **Williams R**, Fernando, B., Lawrence, A., Raji, M., Barclay, J., Brotherson, L., Childs, E., Houghton, J., Khatwa, A., Mills, K., Newton, A., Rockey, F., Rogers, S. L., Souch, C., (2022). The Equator Project – Full Report. https://doi.org/10.31223/X5793T

Published Articles

- Brown, J., **Williams, R.,** Ogburn, S., Brand, B., Breard, E.C.P., Charbonnier, S., Dowey, N., Dufek, J., Jellinek, M., Kueppers, U., Lube, G., Rowley, P., (2026). PDCD-DAT – A global database of pyroclastic density current deposit field data. *Journal of Applied Volcanology* 15, 9. https://doi.org/10.1186/s13617-026-00167-6
- Clare, M.A., Yeo, I.A., Nash, J., Hunt, J.E., Panuve, S., Wilkie, A., **Williams, R**., Dowey, N., Rowley, P., Barclay, J., Phillips, J., Scarlett, J., Engwell, S., Henstock, T.J., Seabrook, S., Watson, S., Wysoczanski, R., Ribo, M., Cronin, S., Talling, P.J., Cassidy, M., Watt, S., Robertson, R., (2025). Volcanic eruptions and the global subsea telecommunications network. *Bulletin of Volcanology, 87, 51*. https://doi.org/10.1007/s00445-025-01832-1
- Rowley, P., **Williams, R**., Johnson, M., Johnston, T., Dowey, N.J., Parsons, D., Provost, A., Roche, O., Smith, G., Walding, N. J., (2025). Chapter 15: Spontaneous unsteadiness and effects of particle size sorting in simulated pyroclastic 1 density currents and the impacts on their deposits. (2025) In *Particulate Gravity Currents: Theory, Experiments, and Environmental Applications*, Geophysical Monograph Series, American Geophysical Union. ISBN: 978-1-394-21671-0 Pre-print at EarthArXiv https://eartharxiv.org/repository/view/5826/
- Walding, N., **Williams, R**., Rowley, P., Dowey, N., Parsons, D. and Bird, A. (2025) Behaviours of pyroclastic and analogue materials, in dry and wet environments, for use in experimental modelling of pyroclastic density currents, *Volcanica*, 8(1), pp 261–285. https://doi.org/10.30909/vol/oxkm9163
- Walding, N., **Williams, R**., Dowey, N, Rowley, P., Thomas. M., Osman, S., Johnson, M., Parsons, D.R. (2025). The influence of moisture on ash strength: implications for understanding volcanic stratigraphy. *Bulletin of Volcanology,* 87, 39. https://doi.org/10.1007/s00445-025-01821-4
- Dowey, N., Lawrence, A., Raji, M., Jackson, C A-L, **Williams R**, Fernando, B., Giles, S., Barclay, J., Brotherson, L., Childs, E., Houghton, J., Khatwa, A., Mills, K., Jameson, G., Rockey, F., Rogers, S. L., Souch, C., (2024). The Equator Project Research School and Mentoring Network: evaluated interventions to improve equity in geoscience research. *Earth Science, Systems and Society,* 4. https://doi.org/10.3389/esss.2024.10123
- **Williams, R**., Mills, K. & Lawrence, A., (2024). Decolonising UK Earth Science. *Geoscientist* 34 (2), 50-53; https://doi.org/10.1144/geosci2024-019
- Walding, N., **Williams, R**., Rowley, P., Dowey, N., (2023). Cohesional behaviours in pyroclastic material and the implications for deposit architecture. *Bulletin of Volcanology*, 85, 67. https://doi.org/10.1007/s00445-023-01682-9
- **Williams, R** & Rowley, P (2023). Anatomy of a volcanic eruption undersea*. Science* 381, 1046-1047. DOI:10.1126/science.adk0181
- Fernando, B., Giles, S., Jackson, C A-L, Lawrence, A., Raji, M., **Williams R**, Barclay, J., Brotherson, L., Childs, E., Houghton, J., Khatwa, A., Mills, K., Newton, A., Rockey, F., Rogers, S. L., Souch, C., Dowey, N., (2023). Strategies for making geoscience PhD recruitment more equitable. *Nature Geoscience.* **16**, 658–660. https://doi.org/10.1038/s41561-023-01241-z
- Raji, M., **Williams, R** & Dowey, N (2023). Widening participation: Lessons from the Equator Research School. *Geoscientis*t, Winter 2022. https://doi.org/10.1144/geosci2022-037
- Dowey, N. & **Williams, R**., (2022). Simultaneous fall and flow during pyroclastic eruptions: A novel proximal hybrid facies. *Geology*; 50 (10): 1187–1191. https://doi.org/10.1130/G50169.1
- Rogers, S.L., Dowey, N., Lau, H., **Williams, R**, (2022). Geology Uprooted! Decolonising the Curriculum for Geologists*. Geoscience Communication*, 5, 189–204. https://doi.org/10.5194/gc-5-189-2022

- Kavanagh, J.L., Annen, C.J., Burchardt, S., Chalk, C., Gallant, E., Morin, J., Scarlett, J., & **Williams, R.** (2022) Volcanologists—who are we and where are we going? *Bulletin of Volcanology* 84, 53 https://doi.org/10.1007/s00445-022-01547-7
- Dowey, N., Barclay, J., Fernando, B., Giles, S., Houghton, J., Jackson, C., Khatwa, A., Lawrence, A., Mills, K., Newton, A., Rogers, S. and **Williams, R**. (2021) A UK perspective on tackling the geoscience racial diversity crisis in the Global North. *Nature Geoscience*, 14(5), 256-259. https://doi.org/10.1038/s41561-021-00737-w Pre-print on EarthArXiv https://doi.org/10.31223/osf.io/z4cju
- Chen, Y., Parsons, D.R., Simmons, S.M., **Williams, R**., Cartigny, M.J.B., Hughes Clarke, J.E., Stacey, C.D., Hage, S., Talling, P.J., Azpiroz-Zabala, M., Clare, M.A., Hizzett, J.L., Heijnen, M.S., Hunt, J.E., Lintern, D.G., Sumner, E.J., Vellinga, A.J. and Vendettuoli, D., (2021), Knickpoints and crescentic bedform interactions in submarine channels. Sedimentology, 68: 1358-1377. https://doi.org/10.1111/sed.12886
- Chevrel, O., Wadsworth, F., Farquharson, J., Kushnir, A., Heap, M., **Williams, R**., Delmelle, P. and Kennedy, B. (2021). Publishing a Special Issue of Reports from the volcano observatories in Latin America: Editorial to Special Issue on Volcano Observatories in Latin America, *Volcanica*, 4(S1), p. i - vi. doi: 10.30909/vol.04.S1.ivi.
- Rotolo, SG.; Scaillet, S.; Speranza, F.; White, J.C; **Williams, R.;** Jordan, N.J., (2021). Volcanological evolution of Pantelleria Island (Strait of Sicily) peralkaline volcano: a review. *Comptes Rendus. Géoscience*, pp. 1-22. doi : 10.5802/crgeos.51.
- Fitton, J.G., **Williams, R**., Saunders, A.D., Barry, T.L., (2020). The role of lithospheric thickness in the formation of ocean islands and seamounts: contrasts between the Louisville and Emperor-Hawaiian hotspot trails. *Journal of Petrology* 61, 11-12 [Gold Open Access] https://doi.org/10.1093/petrology/egaa111
- Geology academic staff at the University of Hull (Bird, A., Caswell, B., Bond, D., Dempsey, E., Dowey, N., Herringshaw, L., Rogerson, M., Widdowson, M., **Williams, R**.) (2020). The Future of Geoscience. *Geoscientist* 30 (8), 8 https://doi.org/doi: 10.1144/geosci2020-103
- *Smith, G., Rowley, P., **Williams, R**. et al., (2020). A bedform phase diagram for dense granular currents. *Nature Communications* 11, 2873. https://doi.org/10.1038/s41467-020-16657-z
- **Williams, R.,** Rowley, P., Garthwaite, M.C., (2019). Small flank failure of Anak Krakatau Volcano caused catastrophic December 2018 Indonesian tsunami. *Geology.* 47 (10): 973–976. doi: https://doi.org/10.1130/G46517.1 Preprint on EarthArXiv doi:10.31223/osf.io/u965c
- **Williams, R**. and Krippner, J. (2019) The use of social media in volcano science communication: challenges and opportunities. *Volcanica*, 1:2, p. i-viii. doi: https://doi.org/10.30909/vol.01.02.i-viii
- Smith, G., **Williams, R**., Rowley, P.J., Parsons, D., (2018). Investigation of variable aeration of monodisperse mixtures: implications for Pyroclastic Density Currents. *Bulletin of Volcanology*, 80:67. https://doi.org/10.1007/s00445-018-1241-1
- Buchs, D.M., **Williams, R.,** Sano, S-I., Wright, V.P., (2018). Non-Hawaiian lithostratigraphy of Louisville seamounts and the formation of guyots. *Journal of Volcanology and Geothermal Research*, 356: 1-23. https://doi.org/10.1016/j.jvolgeores.2017.12.019
- Jordan, N., Rotolo, S., **Williams, R.,** Speranza, F., McIntosh, W., Branney, M., Scaillet, S., (2018). Explosive eruptive history of Pantelleria, Italy: repeated caldera collapse and ignimbrite formation at a peralkaline volcano. *Journal of Volcanology and Geothermal Research*, 349: 47-73. https://doi.org/10.1016/j.jvolgeores.2017.09.013
- Day, S., Walkington, H., **Williams, R.,** Wyse, S., Morton, K., (2016). Shifting landscapes: from coalface to quicksand? Teaching Geography, Earth and Environmental Sciences in Higher Education. *Area*, 48: 308–316. https://doi.org/10.1111/area.12261

- Tejada, M.L.G., Hanyu, T., Ishikawa, A., Senda, R., Suzuki, K., Fitton, J.G., **Williams, R**., (2015). Re-Os isotope and platinum group elements of a FOcal ZOne mantle source, Louisville Seamounts Chain, Pacific Ocean. *Geochemistry, Geophysics, Geosystems*, 16, https://doi.org/10.1002/2014GC005629
- **Williams, R**, Branney, M.J., Barry, T.L., (2014). Temporal and spatial evolution of a waxing then waning catastrophic density current revealed by chemical mapping. *Geology*, 42, 2, 107-110. https://doi.org/10.1130/G34830.1

For full publication list, including prior to 2014, please see my Google Scholar Page.

*WINNER of the Harold Reading Medal in 2020.

Research students

I am the PI of the Catastrophic Flows Research Cluster (6 completed PhD students, 2 current PhD students, 3 completed Masters by Research students). I supervise the following research students:

- 2025 – Present: Benjamin North, PhD, AI and Data Science, University of Hull. Co-I. PI Dr Lara Blythe. Image Analysis and Inverse-Forward Estimation of Pyroclastic Density Currents using Analogue and Process-based Models.
- 2024 – Present: Caitlin Malon, Masters by Research, Geology, University of Hull. Co-I. PI Dr Anna Bird. *Ashy Isotopes: Resolving the Pantescan tephrostratigraphy using trace elements and isotopes*.
- 2023 – Present: Jordan Chenery, PhD, Geology, University of Hull. PI. Co-I Dr Robert Thomas, Dr Natasha Dowey, Dr Pete Rowley, Dr Gareth Keevil. *Can you stop a PDC? Assessing the impact of natural and engineered topographic barriers on pyroclastic density currents.*
- 2022 – 2025: Leah Gingell Masters by Research, Geology, University of Hull. PI. Co-I Dr Xuxu Wu, Dr Pete Rowley, Dr Natasha Dowey. *Is waxing and waning in pyroclastic density currents recorded as reverse grading in an ignimbrite deposit?*
- 2021-2024: Nemi Walding, PhD, Geology & EEI, University of Hull. PI. Co-I Dr Natasha Dowey Dr Pete Rowley, Prof Dan Parsons. *The PDC flow units problem: Deposit heterogeneity from varying cohesive behaviour and sediment flux*
- 2021-2023: Thomas Johnston, Masters by Research, Geology, University of Hull. PI. Co-I Dr Pete Rowley, Dr Natasha Dowey. *How does grain size sorting impact the mobility of aerated granular flows?*
- 2020-2023: Matthew Johnson, Masters by Research, Geology, University of Hull. Co-I PI Dr Natasha Dowey. Co-I Dr Pete Rowley. *The muesli effect in pyroclastic density currents - what does reverse grading in an ignimbrite mean?*
- 2020-2023: Elena Jones, Masters by Research, Geology, University of Hull. Co-I. PI Dr Natasha Dowey, Co-I Dr Lewis Holloway. *Lost in Translation? Exploring the journey from press releases to news articles and mainstream media during volcanic crises, and its impact on public perceptions*
- 2020-2023: Andrew Peck, Masters by Research, Geology, University of Hull. PI. Co-I Dr Janine Krippner, Dr John Whelan. *Who is the hashtag for? An analysis of social media use during the 2017/2018 Agung Volcano Crisis*
- 2016-2021: Chen Ye, PhD, Geology & EEI, University of Hull. Co-I. PI Prof Dan Parsons. *Submarine channel flow dynamics: bedform dynamics and the role of autosuspension*
- 2016-2020: Gregory Smith, PhD , Geology, University of Hull. PI. Co-I Dr Pete Rowley, Prof Dan Parsons. *Propagation of aerated pyroclastic density current analogues: flow behaviour and the formation of bedforms and deposits*
- 2014-2020: Jazmin Scarlett, PhD, Geology, University of Hull. PI. Co-I Prof Greg Bankoff, Dr Briony McDonagh. *Coexisting with volcanoes: the relationships between La Soufriere and the society of St Vincent, Lesser Antilles*
- 2010-2014. Nina Jordan, PhD Student, Department of Geology, University of Leicester. Co-I. PI Prof Mike Branney. *Eruptive history of the peralkaline caldera volcano at Pantelleria, Italy.*

Invited talks

- 2025: Women in Volcanology – who are we, who were we and how do we tell our hidden stories? Yorkshire Geological Society.
- 2024: Deadly volcanic flows: understanding pyroclastic density currents, Geologists Association, online.
- 2023: The land of fire and ice: volcanoes in Iceland. David Rowe Memorial Lecture, Yorkshire Philosophical Society.
- 2023: Can we predict volcanic eruptions? Geography in Action day, Education in Action, London.
- 2023: Past practices in mineral resource and geological exploration and legacy today. Researchers Responding to Volcanic Crises: Uncertainty, Imagination and Improving Best Practice for Collaboration Workshop, London.
- 2023: Quantifying PDCs and the sedimentation of ignimbrites – progress and challenges. Benchmarking and Validating PDC models workshop, IAVCEI, New Zealand.
- 2021: Our Hazardous Earth. Panel Discussion, RGS-IBG online panel series.
- 2019: Erupt: Living with Volcanoes. Cheltenham Science Festival, Cheltenham.
- 2019: The use of social media in volcano science communication - challenges and opportunities. Uppsala University, Sweden.
- 2019: Deadly volcanic flows: understanding pyroclastic density currents. University of Oxford.
- 2018: Deadly clouds and volcanic flows. OpenCampus GEMS, Hull.
- 2018: Deadly clouds and volcanic flows. Yorkshire Philosophical Society, York.
- 2017: Deadly clouds and volcanic flows. Charles Lyell Award Lecture, British Science Festival, Brighton.
- 2017 : Deadly volcanic flows: understanding pyroclastic density currents. Lyell Day Conference, Royal Holloway.
- 2016: Deadly volcanic flows: understanding pyroclastic density currents. University of Durham.
- 2016: Deadly volcanic flows: understanding pyroclastic density currents. Craven & Pendle Geological Society.
- 2016: Lahars triggered by volcanic eruptions. EGU/FORM-OSE Post-graduate Training School 'Landslides and other Geological Hazards in Active Volcanic Environments', São Miguel Island - Azores, Portugal
- 2016 : Deadly volcanic flows: understanding pyroclastic density currents. London Lecture Series, Geological Society of London.
- 2014: A Volcanologist at Sea. Hull Geological Society, Hull.
- 2014: Understanding hazardous volcanic flows. Rotunda Geology Group, Scarborough.
- 2014: Can volcanoes wipe out life on Earth? Cheltenham Science Festival, Cheltenham.

Invited speaker at training or scoping workshops

- 2026 Equator 3.0 Research School, 29th June – 3rd July, Sheffield Hallam University.
- 2023 Equator 2.0 Research School, 3rd – 7th July, University of Birmingham.
- 2023: Panel Member Allyship and supporting others for a more diverse and inclusive geosciences. *EGU General Assembly*. 25th April 2023
- 2022: Equator Research School (co-lead), 8th – 13th April, Sheffield Hallam University.
- 2021: Hidden Histories of Environmental Science Consultation Event. *AHRC-NERC-Collaborative Capacities.* 16th – 17th March 2021
- 2019: Natural hazards: Preparing today to protect tomorrow. *Oxford NERC Environmental Research Doctoral Training Partnership Grand Challenges Seminar Series.* Oxford University 21st May 2019.
- 2016: Lahars triggered by volcanic eruptions. EGU/FORM-OSE Post-graduate Training School 'Landslides and other Geological Hazards in Active Volcanic Environments', July 4-9, Ponta Delgada, Portugal
- 2015: Advancing in Academia, *Staff Development Workshop,* University of Hull, 23rd June 2015.
- 2015: Understanding Volcanic Risk, *STREVA,* University of East Anglia, 8th-9th January 2015.

- 2014: Writing and professional development workshop for teaching-focused GEES academics, *HEA STEM (GEES)*, 22nd-23rd May 2014, Manchester.

Conference convenor

I present my research at international conferences on an annual basis and have been a lead author on 7 and a co-author on >30 international conference abstracts in the last 5 years. Individual abstracts are not listed here.

- 2025: Multidisciplinary Advances in Understanding Pyroclastic Density Currents, IAVCEI Scientific Assembly. Geneva, Switzerland. **Session Convenor**.
- 2024: Particulate Gravity Currents Meeting, University of Hull, UK. **Session Chair.**
- 2023: British Sedimentological Research Group AGM, Loughborough University, UK. **Session Chair**
- 2022: UGUK EDI Monthly Workshop Series. **Series Chair and Organiser**
- 2022: Volcano-sedimentary processes in broader geological environments, EGU Annual Meeting, Vienna, Austria. **Session Convenor**
- 2017: Understanding pyroclastic density currents through analysis of their deposits, IAVCEI Scientific Assembly, Portland, USA. **Session Convenor**
- 2017: Geomorphology and Risk, BSG Annual Conference, Hull. **Session Chair**
- 2016: Volcanism as a hazard and an environmental and climatic agent, VMSG Annual Meeting, Dublin. **Session Chair.**

Editorial positions

- Since 2026: Vice-Editor-in-Chief; DEI and Open Science Officer, Volcanica
- Since 2017: Editorial Board/Committee of Volcanica
- 2017-2022: Advisory Council for EarthArXiv
- 2015 -2019: Editorial board member of Geology

Reviewer for NERC, NSF, Royal Society etc grant proposals and international journal articles.

Professional memberships

- Higher Education Academy; European Geosciences Union; British Society of Geomorphology; Volcanic and Magmatic Studies Group (Committee member 2010-2015); International Association of Volcanology and Chemistry of the Earth's Interior; The Mineralogical Society of Great Britain and Ireland; International Association of Sedimentology

Teaching Experience

National Impact

- Lead of the Decolonising UK Earth Science project https://www.decolearthsci.com/
- Chair of Equality, Diversity and Inclusivity Working Group on the 2021 QAA Earth Science, Environmental Sciences and Environmental Studies Subject Benchmark Statement Advisory Board.
- Head of EDI Portfolio on UGUK Executive Board.

University strategy

University of Hull

- University Student Experience and Success Strategy – Co-Author and co-chair of working group.

- Responsible for Faculty of Science and Engineering Student Experience strategy, working across 7 department.
- Leading on the student experience aspect of Transforming Programmes across FoSE.
- Working with Pro-Vice Chancellor Education, Hull Student Union, Professional Services and Registry
- Member of university-level committees and working groups including Retention Steering Group, Mental Health Working Group, Student Support Task and Finish Group, Sexual violence and domestic abuse Working Group, MEQ Working group, Surveys Group, MyVoice working group, Student Voice Operations Group, Induction & Transition Working Group, Ark Degrees Group, Graduate Outcomes Steering Group

Programme Development

University of Hull

- Led C2016 curriculum design for GEES: Geology Portfolio.
- Major contribution to C2016 curriculum design for GEES: Geography portfolio.
- Developed single honours BSc Geology degree programme (new September 2014, accredited by Geological Society of London January 2016).
- Co-developed joint honours BSc Geology with Physical Geography degree programme (new September 2013, accredited by Geological Society of London 2014).

Module Development

University of Hull

New module developed or significant module redesign.

- Exploring Worlds Around Us (Level 4); Interpreting Environments (Level 4); Geoscience (Level 4 year Maps and structures) now 3D Earth; Geological Materials (Level 4 year igneous and metamorphic rocks) now Rocks, Minerals and Fossils; Volcanoes and their hazards (Level 7) now at Level 6; Basins and Structures (Level 4) now Sedimentary Basins and Structural Analysis; Magmatic and Metamorphic Rocks (Level 5) now Igneous and Metamorphic Geology; Geohazards (Level 5);

Module Leadership and contribution

University of Hull

- Volcanoes and their hazards (module leader 2013-2017; 2021-2026); Professional Consultancy Project (module leader 2023-2026); 3D Earth: Geological Maps and Structure (module leader 2014-2017); Geohazards (module leader 2014-2016); Exploring Worlds Around Us; Interpreting Environments; Rocks, Minerals and Fossils; Landforms and Ecosystems; Sedimentary Basins and Structural Analysis; Geological Mapping Skills; Igneous and Metamorphic Geology; Problem solving and research design; Structural Geology; Dynamic Planet.

FIELD COURSE LEADERSHIP AND CONTRIBUTION

- Introductory geology field courses: Filey (1 day, Level 4); Lake District (5 days, Level 4) - Led the design and delivery; Whitby (1 day, Level 4);
- Interdisciplinary or Physical Geography field courses: Shropshire (3 days, Level 4) - Led the design (~200 students); Swiss Alps field course (7 days; Level 7); Tenerife Field Course (7 days; Level 5)
- Advanced geology field courses: Isle of Skye (11 days, Level 5) – Co-led the design and delivery; Lake District (5 days; Level 5); Almeria, Spain (7 days, Level 5); Anglesey (7 days, Level 5).

Public Engagement

My public engagement work was recognized in 2017 with the British Science Association's Charles Lyell Award Lecture. I have a growing media presence and significant social media presence. I've worked with Sandbox Computers for Kids, Inc, California, USA as a science advisor and mentor. I regularly partake in schools outreach activities and online sessions with classrooms.

TV and Film

- Onscreen expert for NOVA PBS documentary series 'Ancient Earth' 2023
- Onscreen expert for BBC documentary series 'Earth' 2023
- Expert consultant for award-winning, Oscar-nominated film 'Fire of Love' 2022
- Onscreen expert consultant 'Yellowstone Super Volcano' (working title), Wildflame Productions for Discovery and Channel 5 2020.
- Expert consultant for TV e.g. Lion TV Documentary 'The Next Pompeii?'; BBC 2 series, 'Great Continental Railway Journeys'

Books

Author:

- Volcanoes: 10 things you should know. Seven Dials/Orion Publishing, 2025.
- Explanatorium of the Earth, DK Books, Dorling Kindersley Ltd, 2024.

Credited expert consultant:

- Into the Volcano, Flying Eye Books.
- Earth Anthology, DK Books.
- How Everything Works, DK Books.

Media and press coverage

- Regular radio contributor e.g. Good Morning Scotland, 2025; Science slot with Naga Munchetty on BBC Radio 5 Live 2021, 2023, 2025, BBC Radio 5 Live with Tony Livesy, 2025; BBC Radio 5 Live with Colin Murray, 2024; BBC Radio York, 2022, Radio 4 Inside Science 2021; BBC Science in Action 2019; BBC 5Live's Up All Night with Rhod Sharp, 2019; Look North 2018; BBC Coventry 2018; BBC Radio 2 Drivetime with Simon Mayo, March 2017; BBC Radio 2 Breakfast with Chris Evans, April 2017.
- Regular contributor to news articles, National Geographic, etc
- Extensive press coverage of Williams et al., 2019 e.g. https://www.bbc.co.uk/news/av/science-environment-49574670
- The Shift: The Volcanologist, The i100 (The Independent), 3rd October 2015.
- Fuego volcano: the deadly pyroclastic flows that have killed dozens in Guatemala, The Conversation, 4th June 2018.
- Why Japan's deadly Ontake eruption could not be predicted, The Conversation, 30th September 2014.
- Scientists at work: the lava lovers who flock to volcanoes on land and at sea, The Conversation, 28th January 2014.
- Mixing unique fascination with promising career, Hull Daily Mail, 21st January 2014.
- Scientists show how deadly volcanic phenomenon moves, Planet Earth Online, 27th December, 2013. This paper also was covered on numerous science news websites e.g. PhysOrg, Science Daily.
- Erupt from your typical travel pattern, CNN International, 10th July 2013

Other activities and schools/kids clubs outreach

- Cheltenham Science Festival (2014 & 2019); British Science Festival (2017).

- Meet a Volcanologist sessions nationally at schools and youth groups
- “I’m a scientist...get me out of here!” June 2014 winner.
- Nuffield Foundation Placement/Crest Award host and STEM Ambassador
- Expert consultant for Natural History Museum newly redeveloped Earth Galleries.
- Talks available for schools outreach: ‘Can we predict volcanic eruptions?’ ‘Deadly Flows: pyroclastic density currents’.